Anteater Chess User Manual

Version 0.1

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Glossary

Term	Definition
Ant	Another name for a Pawn which is the basic piece that can only move one tile forward unless it has not moved yet. In that case, it would be able to move 2 tiles forward. An Ant is able to promote into any other piece (other than another Ant or King), when it reaches the opposite side of the chessboard.
Ant eating	A special move by the Anteater that can capture multiple Ants at once. The move is initiated by the Anteater capturing the first Ant. If there are any adjacent tiles with Ants, it will target that direction until there are no Ants left, provided that they are at the same rank or file.
Bishop	A piece that moves diagonally across the board any number of spaces.
Castling	A special move that can be played once by each player. This move allows the king to move left or right two tiles to swap places with the Rook on the kingside or queenside. Both the Rook and the King may not have moved a single tile before the attempt to castle. No pieces may be in between them while castling.
Check	When a player's King is vulnerable to capture on the opponent's next turn.
Checkmate	When a player's King is under check and cannot escape.
En Passant	A special pawn capture move that means "in passing". This move starts when an Ant moves two tiles forward from its initial position to attempt to avoid capture from an enemy Ant. The enemy Ant is allowed to capture it as if it had only moved one tile forward.
File	The vertical columns on a chessboard, labeled alphabetically (A-J).

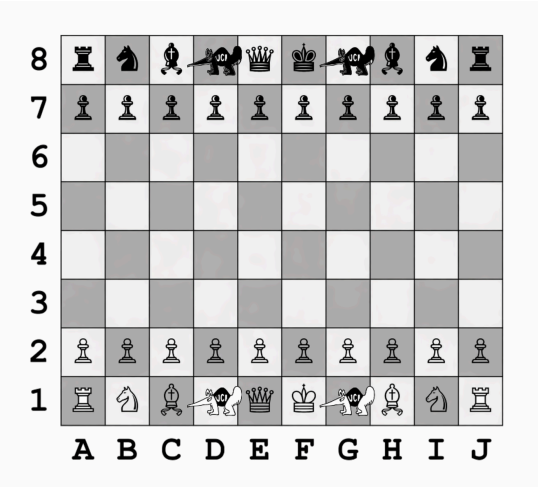
King	The most important piece and target for attack. Moves one square in any direction. The game ends when the king is in checkmate.
Knight	A piece that moves in an L-shape, two squares in one direction and one square perpendicular. The only piece that can jump over other pieces.
Legal moves	The list of moves allowed for a piece. The King's condition is relevant. If a player's king is under check, a move that is played must get the King out of that condition.
Pawn	A piece that moves forward one square, or two squares from its starting position, and may only capture diagonally. Also called an ant.
Promotion	When a pawn reaches the opposite end of the board, it is promoted to another piece
Queen	The most powerful piece, moves in any direction, horizontally, vertically, and diagonally any number of square.
Piece	Items that are moved on a chessboard, each with their unique movesets (King, Queen, Rook, Bishop, Knight, Ant, Anteater).
Rank	The horizontal rows on a chessboard, labeled numerically (1-8).
Rook	A piece that moves horizontally or vertically any number of squares.
Stalemate	When a player's King is not under check, but they are not able to make a legal move, resulting in a draw.

1. Computer Chess

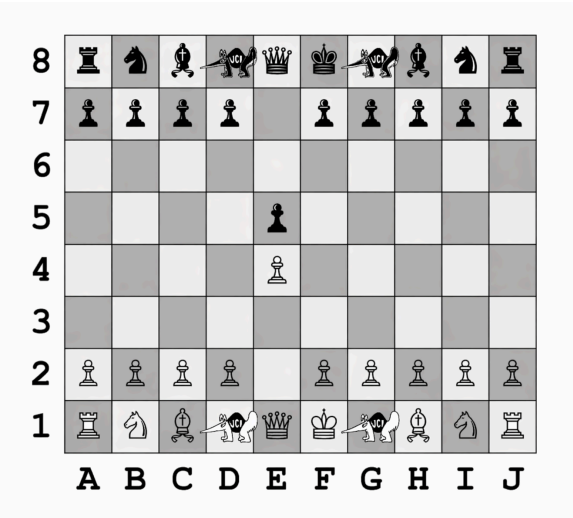
1.1 Usage Scenario

A session begins with the user launching the program from the terminal. The player selects what color pieces to play, and the game is displayed. The players alternate turns and enter moves until the game ends, by checkmate, stalemate, or draw.

Below is the initial position of the game board.



Below is a sample position after the moves E4 E5



1.2 Goals

- Have two players engage in a game of chess.
- Have a player play against a computer where the computer moves in less than 1 minute.
- Allow the player to choose which side of the board to start on.
- Only allow moves that are legal, including special moves (Castling, En Passant, Promotion, Ant eating)
- Log all moves so that a human may read it easily.
- Provide entertainment in a chess game that varies slightly from the original game of chess

1.3 Features

- Allow the player to undo a move.
- Have a player vs. player mode

2. Installation

2.1 System requirements

Operating System: RHEL8 64-bit Linux

2.2 Setup and configuration

1. Download the user package from the server (team16@bondi.eecs.uci.edu/chess)
2. Extract the user package using the following commands:

```
% gtar xvzf BinaryArchive.tar.gz
% evince chess/doc/chess.pdf
% chess/bin/chess
```

- The above commands allow the user to extract the chess folder using [gtar] and view the documentation using the document viewer with [evince]. It will also execute the chess program.

2.3 Uninstalling

1. Execute the following commands:

```
%rm -rf chess
%rm BinaryArchive.tar.gz
```

- The above commands allow the user to remove the extracted chess file, and the user package.

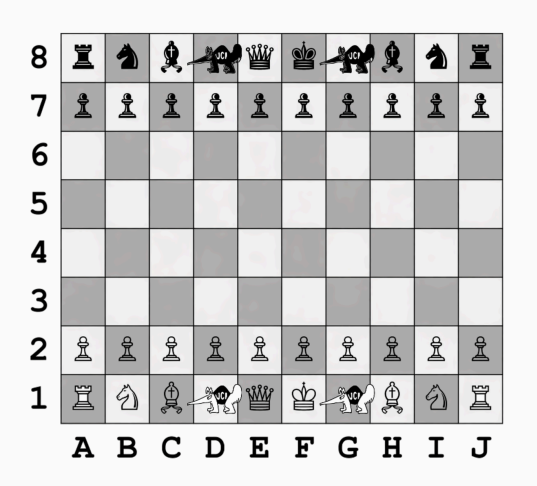
3. Chess Program Functions and Features

3.1 Starting a New Game

When the program launches, a series of buttons appear, then the player is prompted to select to play against a computer or another player. The game will then start unless the player opts to play against a computer in which case the player will be prompted to select what color pieces to play. Once the game is started the game board will appear in its initial position and will begin accepting moves.

User Input: Buttons on the screen to select opponent and piece color.
Program Output: The game board is initialized and the game begins.

Initial Board Position After Launching A New Game



3.2 Making Moves

On a player's turn the program prompts the user for a move. Moves are made by dragging pieces on the game board from their current position to their destination position. If the move is illegal, the game will indicate an error and reject the move, the user will then be prompted to enter another move.

User Input: Dragging pieces on the screen.

Program Output: Confirmation or rejection of the intended move, the game board and log will be updated accordingly.

3.3 The Anteater Piece

The anteater piece is unique to Anteater Chess. Each side starts with two anteaters, one positioned between the king and the king's bishop and the other between the queen and the queen's bishop.

Movement: The anteater moves one square in any direction, similar to the king.

Capture Restrictions: The anteater can only capture pawns (ants). The anteater cannot capture any other piece and does not threaten the opposing king.

Ant eating: The anteater poses a unique ability called ant eating. When a pawn is captured by the anteater, it may continue to capture additional adjacent pawns along the same file or rank in a single move, the player chooses where to stop. The final position is the square of the last pawn captured.

3.4 Game Log

All moves made during the game are recorded using standard chess notation in a log file. The log file displayed on the side of the screen while a game is being played.

Example Log File Being Displayed

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Error Messages

Error	Meaning
Illegal Move. Please try again.	The entered move violates the rules of chess. Check piece movement and game rules and try again.

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